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Scrapie Removal using Planova® Virus Removal Filters

Jun Tateishi[1], Tetsuyuki Kitamoto[2], Shirou Mohri[3], Sakae Satoh[4]*, Tetsuo Sato[4], Ailsa Shepherd[5] and Malcolm R. Macnaughton[5]

[1]Dept. of Neuropathology and [3]Laboratory Animal Center, Kyushu University, Japan; [2]Dept. of Neurological Science, Tohoku University, Japan; [4]Planova Division, Asahi Kasei Corporation, Japan; [5]Biotechnology Dept., Inveresk Research, Tranent, U.K.

Abstract. As a possible method for reducing the risk of transmissible spongiform encephalopathy (TSE) infection, Planova® virus removal filters were tested for their ability to remove scrapie agent ME7. Albumin solution was spiked with high-titre ME7 and filtered through three different pore sizes of Planova® filters. Infectivity of the pre- and post-filtration samples was assayed in log dilutions by intracerebral inoculation into C57B1/6 mice. Filtration of albumin solution in the absence or presence of a detergent (Sarkosyl) with Planova® 35N (35 ± 2 nm mean pore size) removed the contaminating scrapie agent with reduction factors of 4·93 $\log_{10}$ and 1·61 $\log_{10}$, respectively. Filtration, both in the absence and presence of detergent with Planova® 15N (15 ± 2 nm mean pore size), and in the presence of detergent with Planova® 10N (9 ± 2 nm mean pore size), showed high levels of scrapie reduction of >5·87 $\log_{10}$, >4·21 $\log_{10}$, and >3·80 $\log_{10}$, respectively, with no residual infectively detected in any of the filtrate samples. The effectiveness of Planova® 35N filtration for the removal of infectivity of this TSE agent is greatly reduced in the presence of a strong detergent, but Planova® filters with 15 nm or smaller pore size membranes can remove such infectivity at high reduction rates.

Introduction

Bovine spongiform encephalopathy (BSE) and variant Creutzfeldt–Jakob disease (vCJD), occurring mostly in young adults in the U.K., are suspected of being closely related to each other.[1,2] Large amounts of an abnormal isoform of prion protein (PrP^{sc}), which is suspected to be the common pathogen of transmissible spongiform encephalopathies (TSEs) in animals and humans,[3] exist in both diseases. The discovery of a wide distribution of PrP^{sc} in the brain and lymphoreticular tissues, such as the spleen, thymus, tonsils, small intestine wall and other lymph nodes, in vCJD patients has raised concerns regarding the safety of biologicals manufactured from blood and blood components.[4–6]

Epidemiological evidence, though not conclusive, does not suggest that classical or variant CJD is transmitted through blood or blood products.[6] TSE transmission has not been evidenced from patient blood to mice via a peripheral route, the usual entry point in medical practice. However, blood or blood components from mice infected with human and animal TSEs can transmit the disease to other mice via the intracerebral route.[5,7,8]

The triad of defences against infectious risk for manufactured medicinal products consists of source deferral, screening, and inactivation/removal. In the case of the TSE diseases, deferral of source material is greatly limited by the long asymptomatic incubation period, and the screening of raw material is not practical due to the lack of a rapid and sensitive detection method. Since few biological materials can withstand the harsh methods that would be required to assure inactivation of TSE agents, risk reduction from TSE diseases is highly dependent upon the removal of infectivity during production. Trials to remove TSE pathogens from biological materials have been made by several investigators using precipitation,[9] chromatography[10,11] and filtration.[12–14]

In this paper, the Planova® virus removal filter, specifically designed to be applied to the removal of

*To whom correspondence should be addressed: Sakae Satoh, Ph.D., Asahi Kasei Corporation, Planova Division, 9-1 Kanda Mitoshirocho, Chiyoda-ku, Tokyo 101-8481, Japan.

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viruses in the manufacturing process of medicinal products, is tested for its ability to remove scrapie agent ME7, a mouse-adapted strain of scrapie employed as a model for the agents that cause BSE or vCJD.

Materials and methods

Protein solution

Albumin was employed as the protein component of the scrapie removal experiments, and prepared from lyophilized human serum albumin (Catalog No. A-1653, Sigma, St Louis, MO, U.S.A.), which was reconstituted in phosphate-buffered saline (PBS). To examine the effect of detergent on the permeability of the scrapie agent through Planova® filters, albumin solution was prepared containing 0·5% (v/v) Sarkosyl (Sodium N-Lauroylsarcosin, Sigma), an anionic detergent having a chemical formula of $C_{15}H_{28}NO_3Na$, which is known to have a dissociating action for scrapie agent from infected brain tissue, as well as for highly aggregated amyloidic protein. The final concentrations of human albumin with and without Sarkosyl was 2% (w/v). In addition, two scrapie pathogen-free 2% albumin solutions, with and without Sarkosyl, were prepared, and subjected to filtration through Planova® 35N, 15N and 10N filters to examine albumin permeability. The protein concentration was assayed according to the Lowry and Biuret method using a commercially available protein determination kit (Catalog No. 690-A, Sigma).

Filtration

Small-sized (membrane surface area: 0·01 m^2) Planova® filters (Asahi Kasei Corporation, Japan) comprised of a cuprammonium regenerated cellulose membrane in mean pore sizes of 35 ± 2 nm (Planova® 35N), 15 ± 2 nm (Planova® 15N) and 9 ± 2 nm (Planova® 10N) were evaluated. The filtration mode was dead-end at a constant membrane pressure of 0·3 kg/cm^2. The volume of feed solution per membrane surface area was 2 l/m^2. An outline of the filtration steps for the 2% (w/v) human albumin solutions spiked with 1·5% (w/v) scrapie-infected mouse brain homogenate is shown in Figure 1. Air leakage tests before and after filtration as well as gold particle removability tests after filtration were performed to verify the membrane integrity of each filter used.[15]

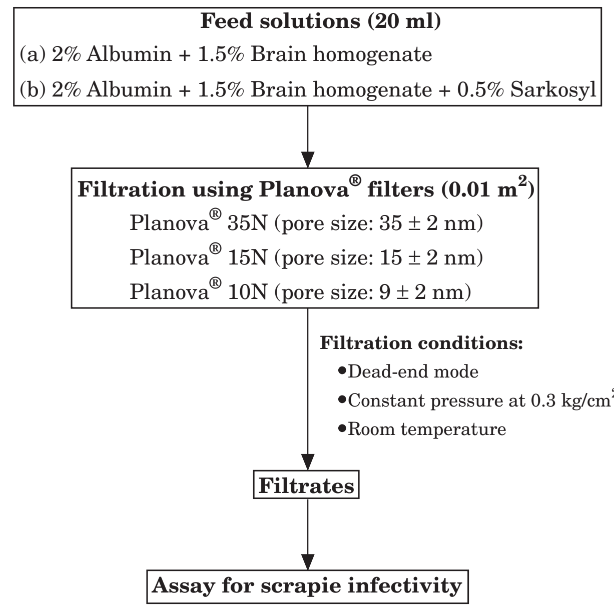

Figure 1. Filtration outline of 2% (w/v) human albumin solution spiked with 1·5% (w/v) scrapie-infected mouse brain homogenate.

Scrapie agent

ME7 is a clonally purified strain of scrapie adapted for C57B1/6 mice that in the presence of the *Sinc s7* genotype gives rise to a short but controlled incubation of murine scrapie. The ME7 agent, which originates from a strain of scrapie found in the Suffolk breed of sheep, was obtained from the BBSRC (Biotechnology and Biological Science Research Council), Edinburgh, U.K. and used to prepare initial scrapie agent stocks. Twelve infected mouse brains were pooled, cooled in an ice bath, and homogenized using a mechanical homogenizer equipped with rotating steel blade, to produce a 30% (w/v) homogenate in saline. Biochemical characterization of the resultant brain homogenate was not conducted. This homogenate was spiked to the previously prepared albumin solution with or without Sarkosyl, and the final concentration of the brain homogenate in the albumin solution was 1·5% (w/v). The spiked solution with Sarkosyl was agitated for at least 1 h at room temperature before filtration.

Animals

C57B1/6 mice were obtained 3–6 weeks of age from Charles River (U.K.) Limited (Kent, U.K.). Animals were acclimatized for a minimum of 4 days before commencement of testing at Inveresk Research (Tranent, U.K.). Ten mice were randomly allocated to each treatment group. Each mouse received a

Table 1. Filtration of scrapie ME7—without detergent

Sample dilution	No. of mice inoculated	No. of mice dying without scrapie	Revised no. of mice used for assessment	No. of mice surviving 20 months without scrapie	No. of mice infected with scrapie	Incubation M ± SD (days)
Feed solution						
10^{-3}	10	0	10	0	10	197 ± 4
10^{-4}	10	1	9	0	9	248 ± 30
10^{-5}	10	0	10	4	6	306 ± 36
10^{-6}	10	4 (4)*	6	6	0	
10^{-7}	9	5 (1)	4	4	0	
10^{-8}	10	4 (3)	6	6	0	
10^{-9}	10	5 (5)	5	5	0	
Filtrate						
Planova® 35N (volume: 19 ml)						
Undiluted	10	3 (2)	7	2	5	320 ± 61
10^{-1}	10	5 (4)	5	5	0	
10^{-2}	10	5 (4)	5	5	0	
Planova® 15N (volume: 22 ml)						
Undiluted	10	4 (2)	6	6	0	
10^{-1}	10	2 (1)	8	8	0	
10^{-2}	9	3 (3)	6	6	0	

*( ): No. of mice dying after mean incubation period plus two SD (378 days) of the most diluted positive control group.

subcutaneously implanted electronic identification chip to identify them individually during the term of the study.

Infectivity assay

Pre- and post-filtration solutions were serially diluted to the appropriate log dilution with saline, and the diluted samples were subjected to scrapie infection assay in C57B1/6 mice. Ten mice were allocated to each assay group. The inoculation size was 0·02 ml per head, and the site of inoculation was in the right parietal cortex of the mouse brain. Unfiltered albumin solution spiked with or without scrapie agent served as positive and negative controls. Mice assigned as test groups were inoculated with the filtered solution. Clinical assessment consisted of mice in each group being checked daily for general health, and a detailed clinical examination for scrapie infection as well as body weight measurement were carried out once a week for a period of up to 20 months. Mice found dead or killed *in extremis* after 100 days post-inoculation or those terminally sacrificed were autopsied for pathological examination. Formalin-fixed mouse brains were cut up into five slices, at least containing thalamus, hippocampus and cerebral cortex, embedded in paraffin, sectioned into 5 μm thicknesses, and stained with haematoxylin and eosin. Immunohistochemical analysis was performed on brain sections of mice in borderline groups where clinical assessment could not be made, and also performed on brain sections of mice in the groups of undiluted and 10^{-1} diluted samples of 15N and 10N filtrates. Polyclonal rabbit antisera raised against a synthetic peptide composed of 25 amino acid residues of the N-terminus of the prion protein[16] was employed for the detection of prion protein. After hydrolytic autoclaving, the prion protein in paraffin-embedded thin sections was stained using the antisera according to the method described by Kitamoto *et al.*[17]

Calculation of reduction factors

Mice not showing clinico-pathological scrapie infection that died before 20 months post-inoculation were not included in the calculation of reduction factors. This was due to the problem of being unable to determine the absence of scrapie infection with early murine mortality. The revised number of mice, instead of the total number inoculated, was used for assessment. However, among the mice dying without scrapie infection listed in Tables 1 and 2, there were many mice that lived longer than the longest incubation period, or mean live day plus two SD (standard deviation), in the positive control groups, and their numbers are shown in parentheses in the same tables. For the

Table 2. Filtration of scrapie ME7—with detergent

Sample dilution	No. of mice inoculated	No. of mice dying without scrapie	Revised no. of mice used for assessment	No. of mice surviving 20 months without scrapie	No. of mice infected with scrapie	Incubation M ± SD (days)
Feed solution						
10^{-3}	10	0	10	0	10	246 ± 18
10^{-4}	10	0	10	6	4	316 ± 46
10^{-5}	10	6 (5)*	4	3	1	335
10^{-6}	10	5 (4)	5	4	1	430
10^{-7}	10	7 (5)	3	3	0	
10^{-8}	10	8 (4)	2	2	0	
10^{-9}	10	6 (4)	4	4	0	
Filtrate						
	Planova® 35N (volume: 22 ml)					
10^{-1}	10	0	10	0	10	226 ± 29
10^{-2}	10	2 (1)	8	0	8	249 ± 31
10^{-3}	10	3 (2)	7	4	3	268 ± 14
10^{-4}	10	4 (4)	6	6	0	
10^{-5}	10	4 (2)	6	6	0	
10^{-6}	10	6 (5)	4	4	0	
10^{-7}	10	5 (4)	5	5	0	
10^{-8}	10	2 (2)	8	8	0	
10^{-9}	10	3 (1)	7	7	0	
	Planova® 15N (volume: 18 ml)					
10^{-1}	10	3 (1)	7	7	0	
10^{-2}	10	5 (4)	5	5	0	
10^{-3}	10	4 (4)	6	6	0	
10^{-4}	10	6 (4)	4	4	0	
10^{-5}	10	1 (1)	9	9	0	
10^{-6}	10	3 (1)	7	7	0	
10^{-7}	10	6 (6)	4	4	0	
10^{-8}	10	3 (2)	7	7	0	
10^{-9}	10	4 (3)	6	6	0	
	Planova® 10N (volume: 20 ml)					
10^{-1}	10	7 (4)	3	3	0	
10^{-2}	10	1 (1)	9	9	0	
10^{-3}	10	7 (6)	3	3	0	
10^{-4}	10	4 (3)	6	6	0	
10^{-5}	10	1 (1)	9	9	0	
10^{-6}	10	7 (5)	3	3	0	
10^{-7}	10	8 (4)	2	2	0	
10^{-8}	10	4 (3)	6	6	0	
10^{-9}	10	3 (3)	7	7	0	

*( ): No. of mice dying after the longest incubation period (430 days) in the positive control group.

purposes of this study, a conservative approach requires that the mice with no pathological changes dying before 20 months post-inoculation not be counted as mice without scrapie infection in the final assessment.

Titres ($\log_{10}$ ID_{50}) of scrapie agent were calculated using the probit statistical analysis outlined by Cavalli-Sforza.[18] The definition of the reduction factor of scrapie infectivity after filtration is analogous to that of virus reduction as described in CPMP guidelines,[19] with results reported in $\log_{10}$ and calculated according to the following equation:

Reduction Factor=

$$\log_{10} \frac{\text{Titre of challenge scrapie spike } (ID_{50}) \times \text{Volume}}{\text{Titre of scrapie recovered in the filtrate } (ID_{50}) \times \text{Volume}}$$

Figure 2. A positive control C57B1/6 mouse inoculated with unfiltered spiking material containing Sarkosyl, at a dilution of 10^{-4}, sacrificed on 306th day. Diffuse deposition of PrP^{sc} in the parietal cortex, hippocampus and the lateral nuclei of the thalamus. Immunostained with anti-PrP.

Results

Negative control mice that had been inoculated with scrapie agent-free albumin solution with or without 0·5% Sarkosyl did not show any signs of scrapie infection (data not shown). Positive control mice which received unfiltered albumin solution containing 1·5% (w/v) brain homogenate with or without 0·5% Sarkosyl developed scrapie with a high transmission rate within relatively short incubation periods (Tables 1 and 2).

Clinical signs of scrapie infection in mice include loss in body weight, piloerection, lethargy, high stepping walk, hunched appearance and wet perigenital area due to urinary incontinence. The severity of these clinical signs corresponded with pathological findings such as spongiform change, astroglial proliferation and loss of nerve cells. Immunohistochemical analysis revealed diffuse deposition of PrP^{sc} in the grey matter, especially in the cerebral cortex, hippocampus and thalamus (Fig. 2). Light microscopy showed PrP^{sc} deposition of various sizes which was more prominent (Fig. 3A) than the spongiform change (Fig. 3B) in the same area. As the cellular isoform of PrP (PrP^{c}) was not stained in the negative control mice by this staining method, a positive stain was considered to be a definite marker of scrapie infection.

In filtration with Planova® 35N filters using 2% albumin solution containing 1·5% (w/v) brain homogenate without Sarkosyl, some scrapie infectivity was observed in the undiluted filtrate, while no infectivity was observed when the filtrate was diluted 10^{-1} by PBS. In filtration with Planova® 15N filters, no scrapie infectivity was observed in either undiluted or diluted samples of filtrate (Table 1).

In filtrations with detergent Sarkosyl, three 2% albumin solutions containing 1·5% (w/v) brain homogenate were each filtered through one of 3 different Planova® filters (Planova® 35N, 15N and 10N), and each resultant filtrate was assayed for infectivity. The scrapie titre in the positive control mice inoculated with protein solution and detergent was slightly smaller than the titre in mice given protein solution without detergent (Table 2). Scrapie infectivity was detected at sample dilutions down to 10^{-3} for Planova® 35N filtrate. In filtration in the presence of Sarkosyl through Planova® 15N and 10N filters, scrapie infectivity was not detected in filtrates at dilutions of 10^{-1} or greater.

Titres of scrapie agent were calculated from the end-point of infectivity using probit statistical analysis. In the case where no infectious agent was observed in the filtrate, the titre was expressed as less than one infectious dose in the volume of the highest concentration tested, which was defined as the minimum detection limit of the assay. For example, in the case of Planova® 15N filtration with detergent in Table 2, calculation of the infectious titre of the filtrate was made as follows. Each mouse was inoculated with a 0·02 ml inoculum of the sample dilution and the total volume of filtrate was 18 ml. (Mice not showing clinical scrapie infection that died before 20 months were not included in the calculation.) Seven mice at 10^{-1} dilution showed no infection, so total volume assayed was:

$$7 \times 0{\cdot}02\ \text{ml} \times 1/10\ (\text{dilution}) = 0{\cdot}014\ \text{ml}.$$

If there was $<1\ ID_{50}$ in 0·014 ml, then

$$18\ \text{ml}/0{\cdot}014\ \text{ml} = 1{\cdot}29 \times 10^{3}\ ID_{50}\ \text{in the filtrate}$$

$$= <3{\cdot}11\ \log_{10}\ \text{of infectivity in the filtrate of Planova® 15N.}$$

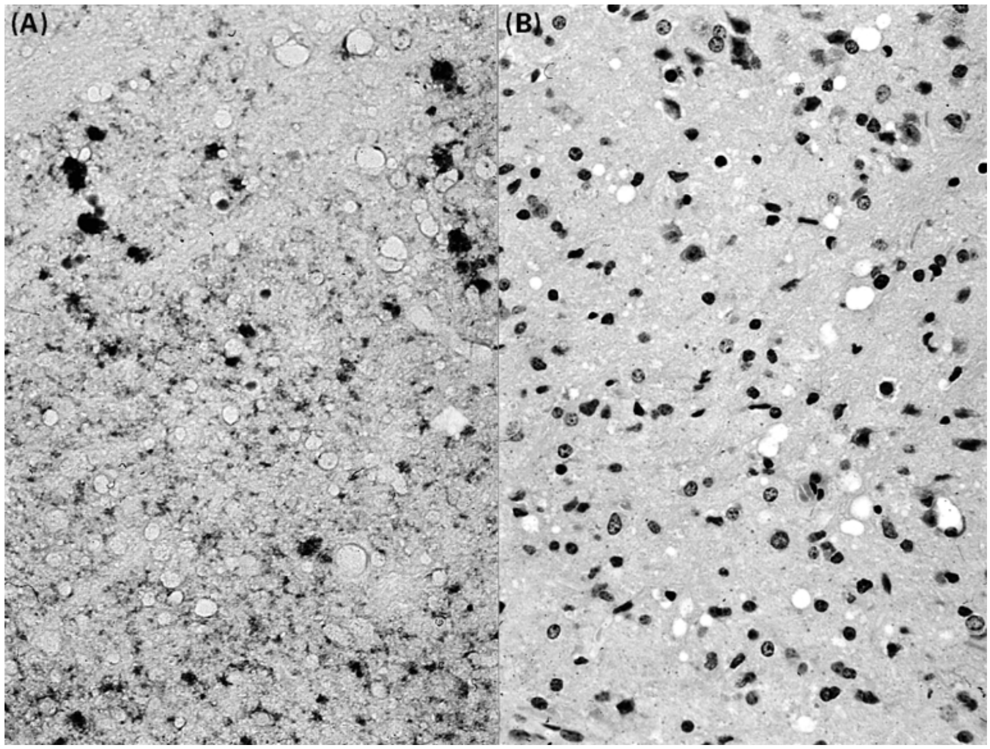

Figure 3. Thalamus of a test mouse inoculated with spiking material filtered through a Planova® 35N filter, undiluted, sacrificed on 385th day. (A) PrP^{sc} deposition of various sizes are more prominent than (B) vacuolation in a neighboring section. A: immunostained with anti-PrP; B: H-E stain.

Table 3. Titre ($\log_{10}$ ID_{50}) and reduction factor (RF)

Sample filtration	Detergent	Titre ($\log_{10}$ ID_{50})	RF*
Feed solution (pre-filtration)	−	8·13	
	+	7·32	
Filtrate (post-filtration)			
Planova® 35N	−	3·20	4·93
	+	5·71	1·61
Planova® 15N	−	<2·26	>5·87
	+	<3·11	>4·21
Planova® 10N	+	<3·52	>3·80

*Reduction Factor (RF)=$\log_{10}$ (Feed solution ID_{50}/Filtrate ID_{50}).

The infectivity titre ($\log_{10}$ ID_{50}/ml) of each feed solution and filtrate along with the corresponding reduction factor (RF) is shown in Table 3. Filtration with or without Sarkosyl using Planova® 35N filters removed the spiked scrapie agent with reduction factors of 4·93 $\log_{10}$ and 1·61 $\log_{10}$, respectively. Filtration, in the absence or presence of Sarkosyl with Planova® 15N filters, and in the presence of Sarkosyl with the Planova® 10N filter, showed high levels of scrapie reduction at >5·87 $\log_{10}$, >4·21 $\log_{10}$ and >3·80 $\log_{10}$, respectively, with no residual infectivity detected in any of the samples.

All filters used in this study were subjected to integrity testing (air leakage test and gold particle removability test). All filters conformed to pre-established specifications (data not shown).

Protein permeability tests were performed independently using Planova® 35N, 15N and 10N filters.

Table 4. Protein recovery after filtration

Filter	Mean pore size (nm)*	Detergent†, presence/absence	Protein recovery‡ (%)
Planova® 35N	35 ± 2	Absence	79·78
		Presence	88·43
Planova® 15N	15 ± 2	Absence	70·57
		Presence	67·65
Planova® 10N	9 ± 2	Absence	74·03
		Presence	65·71

*Mean pore size is determined by the water flow rate method.
†0·5% (v/v) Sarkosyl.
‡2% (w/v) albumin.

Two 20-ml solutions of 2% albumin, one of them containing 0·5% Sarkosyl, were filtered through Planova® 35N, 15N and 10N filters, and the protein concentrations before and after filtration were assayed using a Sigma protein determination kit. The results are shown in Table 4. The albumin recovery rate was greater in the presence of Sarkosyl for the Planova® 35N filter than in the detergent's absence, but the converse was true for Planova® 15N and 10N filters where the protein recovery rate was lower in the presence of Sarkosyl than in its absence.

Discussion

It is widely understood that the causative agent of TSEs has a strong correlation to the presence of the isoform (PrP^{sc}) of normal prion protein (PrP^{c}). The amino acid sequence of the PrP^{sc} molecule has been determined, and the relationship between the type of TSE disease and its protein structure has been studied extensively.[20,21]

To prevent widespread TSE transmission via biological materials for human use, various TSE agent inactivation methods have been examined by a number of investigators who found that physical and chemical inactivation of TSE agents is extremely difficult.[22,23] Methods known to inactivate TSE agents are too harsh to be used to treat biologically fragile active substances like blood-derived products and cell culture products. As a consequence, the available methods for risk reduction from TSE transmission through biologicals are limited, with the only practical solutions considered to be separation by chromatography or removal by filtration.

Two of the authors (J.T. and T.K.) have attempted to remove human TSE pathogen from diluted brain homogenate by passing the homogenate through a screen-type membrane filter (Millipore, Bedford, MA, U.S.A.) with a pore size of 0·025 μm, and initially found that the TSE pathogen was removable by filtration.[12] However, in that trial, only a small amount of diluted brain homogenate could pass through the membrane even though filtration was conducted under high pressure. The two authors separately reported that a large amount of brain homogenate, prepared from mice infected with human TSE, Fukuoka-1 strain, was successfully filtered through a microporous membrane filter, the Planova® virus removal filter, with membrane mean pore sizes ranging from 75 nm to 10 nm.[13] No infectivity was observed in the filtrates from the Planova® filters with a pore size of 35 nm. Infectivity was observed in the filtrate where the feed TSE solution containing 1% Sarkosyl was filtered through a 10 nm pore size Planova® membrane.[14]

The present experiment using scrapie ME7 strain produced additional data on TSE filtration, showing variability in the reduction of TSE pathogens by filtration. These differences may have resulted from a number of factors including differences in the source pathogens used (ME7 *vs*. Fukuoka-1), the presence or absence of human albumin in the feed protein solutions, and the use of different mouse strains (C57B1/6 *vs*. NZW). From the results observed in these two filtration studies, the following suppositions can be put forth for further investigation.

(a) In the absence of Sarkosyl detergent, some infectivity of scrapie ME7 was detected in the filtrate of the 35 nm pore membrane, while infectivity of Fukuoka-1 did not appear in the filtrate of the same pore size membrane. These findings may suggest that the binding of the ME7 infectious agent (PrP^{sc} in aggregate form) to tissue components is not so strong, allowing released aggregates to pass through the 35 nm pore membrane, while the binding of Fukuoka-1 aggregates to tissue components is much stronger, resulting in all infectious agents being trapped within the 35 nm pore membrane.

(b) In the presence of 0·5% Sarkosyl where some self disaggregation or dissociation of PrP^{sc} aggregates from tissue components is expected, a large amount of ME7 infectivity appeared in the filtrate of the 35 nm pore membrane, while no infectivity was observed in the filtrate of 15 nm or smaller pore membranes. However, in the previous filtration study that used the Fukuoka-1 strain in the presence of 1% Sarkosyl, some infectivity did pass through the 10 nm membrane.[14] In these two studies, the concentrations of detergent Sarkosyl were different, 0·5% for ME7 and 1% for Fukuoka-1, but, as Sarkosyl is one of the strongest anionic detergents known, it can be presumed that the efficiency of Sarkosyl for protein dissociation is not significantly affected by the difference between the 0·5% and 1% concentrations. Accordingly, it can further be presumed that most of the tissue-bound PrP^{sc} aggregates were released in the presence of Sarkosyl detergent. Based on the above presumptions, the results of these two filtration studies may suggest that the size of the infectious agent (PrP^{sc} aggregate) of ME7 formed in C57B1/6 mouse is larger than that of Fukuoka-1 formed in NZW mouse.

(c) It can further be inferred that the molecular conformation of PrP^{sc} formed by scrapie induced ME7 in C57B1/6 mouse differs from that of PrP^{sc} formed by human TSE-induced Fukuoka-1 in NZW mouse. This conformation difference may result in the formation of aggregates which, as a consequence, display different behavior patterns to Sarkosyl treatment and the subsequent filtration. Such conformation difference seems plausible in that human PrP^{sc} is classified into two types, Type-1 and Type-2, on the basis of different susceptible sites to proteinase K, even though the amino acid sequence of PrP is completely the same in both types, indicating the existence of two different conformations of PrP^{sc} aggregate in human TSE.[24] This is strong evidence which supports the above speculation that conformation of aggregated PrP^{sc} in mouse varies according to the TSE species used for induction of the disease.

The protein recovery data shown in Table 4 contradict the expected recovery rates after filtration of greater than 95% with 35 nm and 15 nm pore size membranes, as the molecular weight of albumin is around 60 kDa. These low recovery rates are attributed to the purity of the lyophilized albumin powder used. Gel permeation chromatography produced an elution chart of the lyophilized albumin showing a broad shoulder just before elution of the main albumin peak, with an intensity calculated to be 20–25% of the total absorption at 280 nm (data not shown). This shoulder indicates the presence of some substances larger than albumin within the lyophilized albumin powder, which may have been filtered and removed by the Planova® filters. Taking into account the existence these substances in the feed solution, the actual recovery rate of albumin is estimated to be greater than 90% for each filter type.

The results obtained in this study demonstrate that the effectiveness of a 35 nm membrane (Planova® 35N) to remove scrapie-induced ME7 infectivity assayed in a C57B1/6 system is greatly reduced in the presence of a strong detergent, but that a 15 nm membrane (Planova® 15N) or smaller pore membrane can remove such infectivity at high reduction rates.

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